# 11. Certified topological bands in an idealized Maxwell medium

Pure Thought research portfolio

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**Revised specification — 2026-09-11.** Detailed critique · Original description · Evidence standard

## Plain-language guide

### The problem in everyday terms

A repeating optical structure changes which light waves can travel through it. With suitable nonreciprocal response, it may also support light that travels along an interface in a preferred direction. This problem asks whether a precisely defined ideal electromagnetic medium really has the required band structure and topology. It separates a mathematically guaranteed wave effect from the later task of finding a material that realizes the assumed response.

### Key terms

- **Maxwell equations** — The equations governing classical electric and magnetic fields.
- **Periodic medium** — A medium whose properties repeat from cell to cell.
- **Permittivity $\epsilon$ and permeability $\mu$** — Quantities describing how a medium responds to electric and magnetic fields; they can be direction-dependent tensors.
- **Nonreciprocity** — Response that breaks the usual interchange symmetry between source and receiver; here it provides the required time-reversal-breaking mechanism.
- **Photonic band** — An allowed family of electromagnetic wave frequencies across crystal momenta.
- **Transverse constraint** — The condition selecting physical wave fields consistent with the relevant divergence equation.
- **Chern number** — An integer characterizing the topology of an isolated group of wave bands.
- **Interface mode** — A wave concentrated near the boundary between two media.
- **Lossless and nondispersive** — Idealizations in which energy is not absorbed and the constitutive response is taken independent of frequency.

### Why this matters

Controlling where and in which direction waves travel is a basic aim of optical engineering. Topology offers one mechanism for robust interface behavior under appropriate conditions. A certified Maxwell calculation would also expose whether an apparent effect is caused by numerical artifacts or by assumptions a real material cannot satisfy.

### What progress would mean

The first meaningful achievement would be a validated ideal medium with a proved gap and topological index. It could inform later waveguide designs, but performance in a lossy, frequency-dependent device would require a separate physical model.

## Core question

Can a specified lossless periodic electromagnetic model support a rigorously isolated band group with nonzero Chern number, and can its interface modes be validated?

## Scope and assumptions

Choose a two-dimensional periodic, lossless, frequency-independent constitutive model with Hermitian positive $\epsilon(\mathrm{r})$, $\mu(\mathrm{r})$, specified boundary conditions, and a mechanism breaking physical time reversal. Treat the constitutive tensors as mathematical inputs. Real-frequency dispersion, loss, and fabrication feasibility are extensions requiring additional physical data.

Start by validating a reciprocal scalar dielectric benchmark with zero total Chern number for a time-reversal-invariant band group; then introduce an explicitly nonreciprocal tensor.

## Mathematical target

In a formulation supporting the chosen model, solve

$$(\nabla + ik) \times \mu^{-1}(r)(\nabla + ik) \times u = \omega^2 \epsilon(r) u,$$

in units with c=1, subject to the appropriate transverse constraint. Use the $\epsilon$-weighted inner product or an equivalent Hermitian formulation. Specify an isolated positive-frequency band group; photons do not require an electron-like filling assumption. Prove a frequency gap over the entire Brillouin zone and certify the projector invariant in the correct metric.

## Required outputs and proof obligations

Return ideal geometry and constitutive tensors, solver discretization, eigenvalue enclosures, spurious-mode controls, gap and topology certificates. Interface calculations must state both media and termination. Report robustness only for an explicit perturbation class and bound that preserves the required gap/symmetry.

## Validation and rejection controls

Check homogeneous-medium dispersion, mesh/basis convergence, Hermiticity, and transverse modes. Separate Galerkin error from k-space interpolation error. Verify reciprocal controls cannot acquire a net Chern number from numerical artifacts. Compare an interface spectrum with the bulk-index difference without claiming immunity to arbitrary disorder.

## Milestones and research extension

Baseline: validated reciprocal eigenproblem. Strong: one certified ideal nonreciprocal topological design. Research extension: a realistic dispersive constitutive model or gap optimization within a bounded geometry family. Fabrication drawings alone do not certify material behavior.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

## Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

## Primary references and their role

- Haldane and Raghu, Directional optical waveguides with broken time-reversal symmetry — Nonreciprocal media as the mechanism for photonic quantum-Hall-like edge modes.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

## Displayed formulation and claim boundary

Canonical specification.

$$(\nabla + ik) \times \mu^{-1}(r)(\nabla + ik) \times u = \omega^2 \epsilon(r) u, \qquad \langle u, v \rangle_\epsilon = \int_{\text{cell}} u^* \cdot \epsilon v \, dr.$$

Use positive Hermitian constitutive tensors, the transverse constraint, units $c = 1$, and an isolated positive-frequency band group. Certify its gap and projector invariant in the weighted metric. Nonzero total Chern number requires breaking the relevant physical time-reversal symmetry; scalar reciprocal benchmarks provide rejection controls.

## Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

### Detailed findings

1. **Blocking — missing nonreciprocal constitutive physics (68–98).** A real scalar dielectric model does not implement the claimed gyromagnetic Chern phase. The relevant tensor and inner product must be included.
2. **Major — electronic $\mathbf{Z}_2$ shortcut (95–98).** Ordinary photonic time reversal squares to $+1$; an electronic parity formula cannot be transplanted without an appropriate additional symmetry structure.
3. **Major — "exact" Maxwell solution (21–25, 56–60).** Exact governing equations do not imply exact numerical eigenvalues. Discretization, spurious longitudinal modes, and continuum coverage need validation.
4. **Major — fabrication and robustness (37–42, 740–754).** Generic dielectric rods and STL files do not establish nonreciprocity, telecom performance, or a universal 15% disorder

tolerance. The new problem certifies a mathematical medium first and identifies the extra evidence required for engineering claims.

### What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Can a specified lossless periodic electromagnetic model support a rigorously isolated band group with nonzero Chern number, and can its interface modes be validated?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

### Acceptance criteria to use instead

Baseline: validated reciprocal eigenproblem. Strong: one certified ideal nonreciprocal topological design. Research extension: a realistic dispersive constitutive model or gap optimization within a bounded geometry family. Fabrication drawings alone do not certify material behavior.

### Sources supporting the corrections

- Haldane and Raghu, Directional optical waveguides with broken time-reversal symmetry — Nonreciprocal media as the mechanism for photonic quantum-Hall-like edge modes.

## Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

### What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

### Four evidence levels

| Label | What it establishes | What it does not establish |
|---|---|---|
| Numerical evidence | A reproducible computation at stated precision, sample size, and discretization | Exact truth, continuum coverage, or a universal theorem |
| Finite-model certificate | An exact/validated statement about the specified finite model or finite relaxation | An omitted tail, thermodynamic limit, or physical realization |
| Conditional mathematical theorem | A proof over the stated domain with every hypothesis and error controlled | Truth outside those assumptions or a demonstrated physical system |
| Formal proof | A checked theorem in a named proof assistant, with imported axioms/dependencies listed | Correct physical modeling merely by virtue of formalization |

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

### Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

### Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.

- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

### Checking specific certificate types

**LP/SDP/SOS:** verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

**Interval numerics:** state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

**SAT/SMT:** a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

**Statistics:** specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

### Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.